CUSTOMER CHURN ANALYSIS

Project Documentation • Python • Pandas • SQL/SQLite • Data Analysis

A portfolio-ready overview of the project objectives, calculated metrics, analysis workflow, findings template, and potential business recommendations.

Prepared for GitHub portfolio documentation

1. Project Overview

This project analyses customer churn for a subscription-based business to understand customer attrition, identify higher-risk customer segments, and explore factors that may be associated with churn. The workflow uses Python, Pandas, SQL/SQLite, and data visualisation to turn customer, subscription, and support data into actionable findings.

Project Objectives

• Measure overall customer churn and retention.

• Compare churn across plan types, contract types, and states.

• Analyse customer tenure and monthly charges.

• Segment customers by churn risk.

• Investigate the relationship between support escalations and churn.

• Estimate monthly charges associated with customers who may be at risk.

• Translate analysis into practical retention recommendations.

2. Tools and Technologies

• **Python:** data processing and analysis

• **Pandas:** cleaning, transformation, grouping, and aggregation

• **SQLite / SQL:** querying customer and support data

• **Jupyter Notebook:** analysis workflow and reproducibility

• **Data visualisation:** charts and summary reporting

3. Data Preparation

The workflow connects to a SQLite database, loads customer and support-related tables into Pandas, reviews data types and missing values, checks duplicates, transforms fields, and creates derived variables for analysis.

Churn-Risk Segmentation

Churn score	Risk category
Below 50	Low
50 to below 70	Medium
70 or above	High

Confirm these thresholds match the final version of the notebook before publishing the project.

4. Calculated Metrics

KPI	Formula / Definition	Purpose
Churn Rate	Churned customers ÷ total customers × 100	Share of customers who churned.
Retention Rate	100% – churn rate	Share retained, using the same population and period.
Churn by Plan Type	Churn rate calculated for each plan type	Compare churn across plans.
Churn by State	Churn rate calculated for each state	Identify geographic differences.
Churn by Contract Type	Churn rate calculated for each contract type	Compare contract groups.

KPI	Formula / Definition	Purpose
ARPU	Monthly revenue from active customers ÷ active customers	Average monthly revenue per active customer.
Average Customer Tenure	Average customer tenure in days	Describe how long customers remain subscribed.
Revenue at Risk	Sum of monthly charges for high-risk customers	Estimate monthly charges associated with high-risk customers.
Escalation Rate	Customers with ≥1 escalation ÷ total customers × 100	Customer-level escalation prevalence.
Average Complaints per Customer	Total complaints ÷ distinct customers	Average complaint volume per customer, if complaint data exists.
Churn by Escalation Status	Churn rate with escalations vs. without escalations	Explore association between support escalations and churn.

Metric notes: Keep the analysis period and customer population consistent. If escalations are stored as counts, distinguish counts from the percentage of customers with at least one escalation. Revenue at risk is not necessarily realised revenue loss. Correlation does not prove causation.

5. Analysis Approach

Overall Churn and Retention

Calculate churn and retention rates to establish a baseline.

Plan and Contract Analysis

Group by plan and contract type, comparing customer counts and churn rates.

Geographic Analysis

Calculate churn by state and consider customer counts before interpreting outliers.

Tenure and Revenue Analysis

Calculate tenure from subscription start and cancellation dates where available; compare customer groups.

Risk Segmentation

Group customers into Low, Medium, and High risk and compare counts, churn behaviour, and monthly charges.

Support and Escalation Analysis

Join customer and support data using the correct customer identifier and compare churn by escalation status.

6. Insights and Findings

Fill in this section with verified outputs from your own notebook. Do not present values from reference material as results from this dataset.

Business question	Result to add
Overall churn rate	Add calculated percentage
Plan type with highest churn	Add plan name and churn rate
State with highest churn	Add state, churn rate, and customer count
Contract type with highest churn	Add contract type and churn rate
Average customer tenure	Add average tenure and unit
Customers by risk category	Add Low, Medium, and High risk counts

Business question	Result to add
Monthly charges associated with high-risk customers	Add amount and currency
Churn by escalation status	Compare both churn rates and group sizes

7. Potential Business Recommendations

• Review pricing, service quality, and customer feedback in segments with elevated churn.

• Prioritise outreach to high-risk customers where customer value and risk justify the effort.

• Investigate whether support escalations are associated with higher churn.

• Review contract and plan differences before proposing contract-migration or retention offers.

• Monitor churn and retention KPIs over time to evaluate whether interventions are effective.

These are potential actions to test, not claims that the analysis has already proven their effectiveness.

8. How to Run the Project

• Clone or download the repository.

• Install required libraries, for example: `pip install pandas`. Install any additional libraries used in your notebook.

• Ensure the source database is in the expected location, or update the database path in the notebook.

• Open the Jupyter Notebook and run cells in order.

• Update the findings section with verified results from the notebook.

Privacy: Do not upload confidential or personally identifiable customer information to GitHub. Exclude sensitive database files and document how to provide them locally.

9. Project Summary

Customer Churn Analysis is a data analytics project focused on understanding subscription churn, customer risk, revenue exposure, and support-related patterns. Using Python, Pandas, and SQL/SQLite, the workflow prepares customer data, calculates retention-related KPIs, segments customers by churn risk, and compares churn across customer groups.

The project is intended to support evidence-based retention decisions by highlighting customer segments that may require further investigation. Final business conclusions and recommendations should be based on verified results produced by the project notebook.